

Medicine Outcome Prediction System

Model Comparison and Selection Report

Recommended Model

XGBoost

Selection Basis:

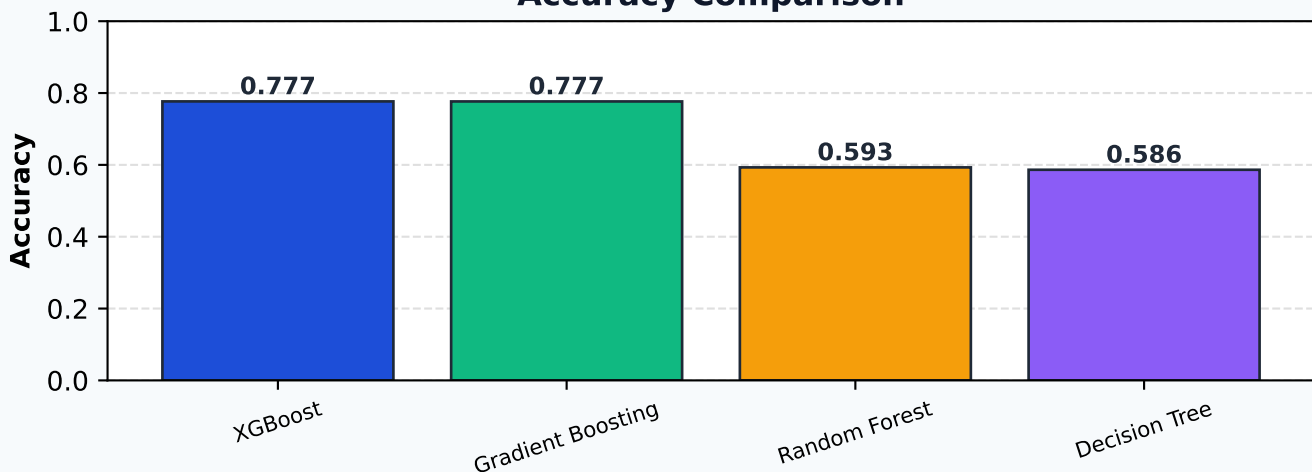
Accuracy: 0.7767

Precision: 0.5000 Recall: 0.0100 F1-score: 0.0100

Data split: 80% Training / 20% Testing (stratified)

Evaluation metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix

Accuracy Comparison



Detailed Model Evaluation

Model	Accuracy	Precision	Recall	F1-score	Confusion Matrix
XGBoost	0.7767	0.5000	0.0100	0.0100	[[152837, 300] [43735, 304]]
Gradient Boosting	0.7766	0.0000	0.0000	0.0000	[[153136, 1] [44039, 0]]
Random Forest	0.5930	0.3000	0.6300	0.4100	[[89369, 63768] [16484, 27555]]
Decision Tree	0.5864	0.3000	0.6400	0.4100	[[87456, 65681] [15862, 28177]]

Interpretation:

XGBoost is the strongest model in terms of overall predictive accuracy for this clinical outcome data.

For a clinical use case, recall and F1-score for the positive outcome should also be reviewed alongside a

Confusion Matrix Comparison

